Cartelli di Disfida matematica

|p.25| I am pleased, Messer Nicolò, that through these questions you have given me the opportunity to help those who take delight in Geometry and Arithmetic, although they have not yet reached the highest level of these sciences. Your first seventeen questions concern that remarkable method of working without changing the opening of the compass. I do not know who first devised it. I do know, however, that over the last fifty years many brilliant minds have devoted themselves to improving it. Among them, the late Messer Scipione dal Ferro of Bologna deserves special remembrance.

My aim, therefore, is to bring this method to its full potential. By means of it, I shall prove not only a number of propositions discovered by our predecessors, but the whole of Euclid.

Your final questions have given me a second opportunity to be of service, this time to students of arithmetic. [...]

The aim is to prove the whole of Euclid without changing the opening of the compass. In place of the following postulate,

> *Super quovis centro, quantum libet occupando spatium, circulum describere.*
>
> To describe a circle with any given center and any desired radius.

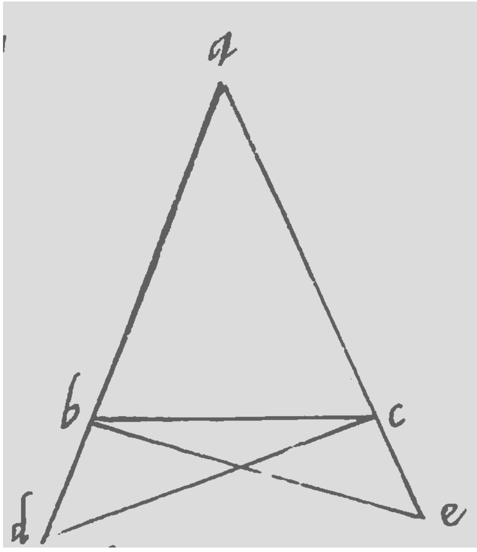
I shall adopt another:

> To draw a circle about any given centre with whatever compass opening |p.26| has been assigned by one's opponent.

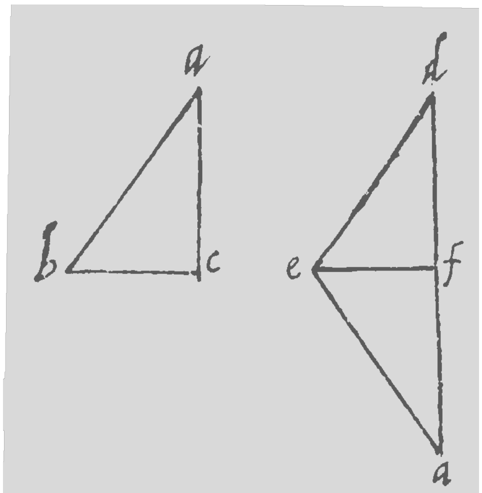
The first proposition I shall prove is Proposition I.4. I prove it exactly as Theon does, since it requires neither the preceding propositions nor the omitted postulate. The second will be Proposition I.5. Since I cannot make use of the third postulate, I proceed as follows.

Let ABC be a triangle with equal sides AB and AC. By the second postulate I extend AB and AC indefinitely. Then, taking B and C as centres, with the prescribed compass opening proposed by my opponent. I cut off the segment BD from the extended line AB, and the segment CE from the extended line AC. Since the lines AB and AC are equal, and since BD and CE are also equal, each being equal to the same compass opening, it follows, by the second common notion, that the lines AD and AE are equal.

Next I draw the segments DC and BE, and complete the proof as Theon does.

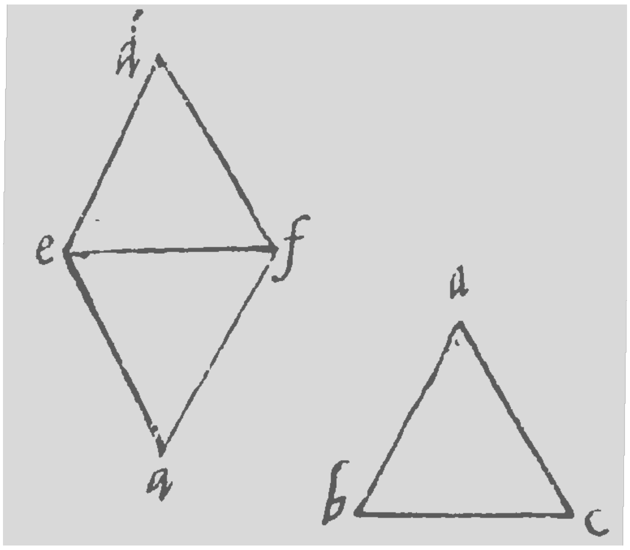


The third proposition will be Proposition I.8. I shall prove it by direct demonstration (*dimostrazione ostensiva*), making use of no proposition other than the two already established. I have taken this proof from Proclus, in the third book of his commentary on the First Book of Euclid. Let there be two triangles, ABC and DEF, satisfying the conditions of Proposition I.8: let AB be equal to DE, AC equal to DF, and BC equal to EF. Following the method used in Proposition I.4, I superimpose the segment BC upon EF, placing the point A on the lower side, so that the triangle AEF coincides with the triangle ABC. I then draw the segment DA. It must fall on the line DF, either inside or outside. Let us first suppose that it falls upon DF, as in the first figure.



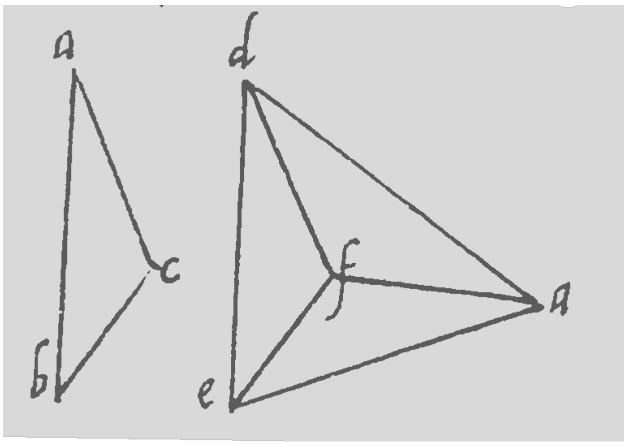
It must therefore also pass through the point F; otherwise two straight lines would enclose an area, contrary to the last postulate. Since the lines ED and EA are equal, it follows, by the second proposition, that the angles EAD and E/p.27/DA are equal, which is what had to be proved.

If the line falls inside, as in the second figure,



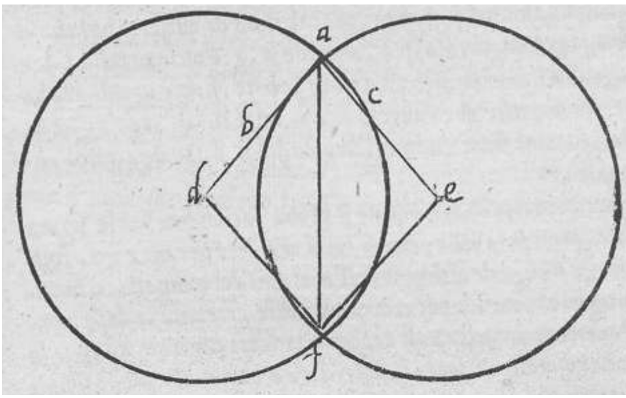
likewise, by the second proposition, the angle FDA is equal to the angle FAD, and the angle EDA to the angle EAD. Hence, by the second common notion, the whole angle D is equal to the whole angle A, which is what was to be proved.

If, on the other hand, it falls outside, as in the third figure,



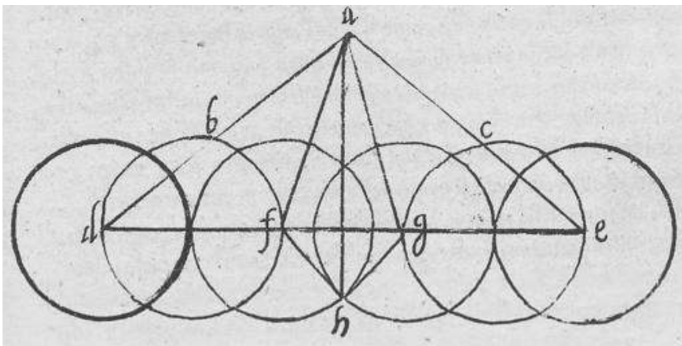
again, by the second proposition, the angle EDA is equal to the angle EAD, and likewise the angle FDA is equal to the angle FAD. Therefore, by the third common notion, the angle EDF is equal to the angle EAF, that is, to the angle BAC, which is what was to be proved.

The fourth of these will be Proposition I.9. Let the angle to be bisected be the angle BAC.



By the second postulate, let the lines AB and AC be extended indefinitely. Taking A as the centre, and using the compass opening assigned by my opponent, I draw a circle which, for example, intersects the extended lines AB and AC at the points D and E. Next, with the same compass opening, I draw two circles centred at D and E. Let them intersect at the points A and F (as they must, since both AD and AE are equal to the given compass opening). Next, I draw the segment AF, which, I claim, bisects the angle BAC.

To prove this, I draw the segments DF and EF, thus obtaining the triangles ADF and AEF. Since the sides DA and AF of the first are equal to the sides EA and AF of the second, and since the base DF is equal to the |p.28| the base FE, by the third proposition, it follows that the angle DAF is equal to the angle EAF, which is what was to be proved. Even if the equal sides AD and AE were so long, and so positioned, that the line DE were greater than twice the compass opening, the same conclusion could still be reached by constructing, as shown in the figure,



as many circles as needed on either side until, at last, they either touched at a single point or intersected, as do in the figure the two circles centred at F and G. If they happened merely to touch, then the line drawn from the point of tangency to the point A would bisect the angle by Proposition III.

If, instead, they intersect, as shown in the figure at the point H, then the segment HA is the required angle bisector.

To prove this, draw the four segments AF, AG, FH, and GH. Since AE is equal to AD by the second proposition, angle D is equal to angle E. Since EG is equal to DF, the first proposition implies that the base AG is equal to the base AF, and that the angle DAF is equal to the angle EAG. Next, since the sides FA and AH are equal to the sides GA and AH, respectively, and the base FH is equal to the base GH, Proposition III implies that the angle FAH is equal to the angle GAH. Hence, by the second common notion, the whole angle DAH is equal to the whole angle EAH, which is what was to be proved.

The fifth of these will be Proposition I.10, which is easily obtained from the figures of the preceding proposition. Suppose that the line to be bisected is DE, which must be either equal to, shorter than, or longer than twice the given compass opening. If it is equal to it, then, taking one endpoint as the centre and drawing a circle with the prescribed compass opening, the circle will bisect the line. If it is shorter, then, drawing two circles with centres D and E, as in the first figure of the preceding proposition, they will intersect at the two points A and F; and the segment AF will bisect the line DE, because, by bisecting the angle, it also bisects the line by the first proposition, forming the two triangles and arguing as Theon does.

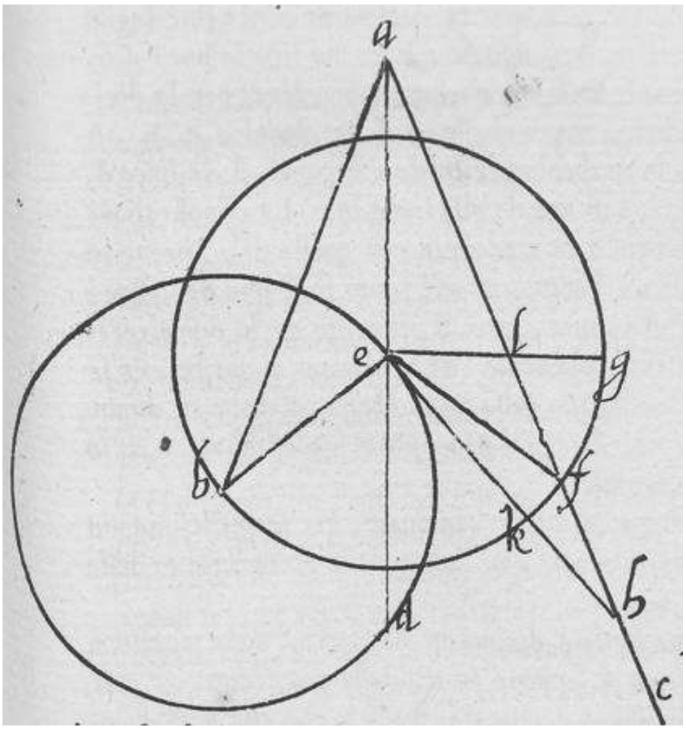
If, however, it is longer, I proceed as in the second figure of the preceding proposition. From one intersection to the other of the two circles drawn with centres F and G, I draw a line which, just as it bisects the angle, also bisects the line FG (arguing by the first proposition). Now the line FD is equal to the line GE. Therefore, by the second common notion, the whole line DE is thereby bisected.

Finally, if the two circles drawn with centres F and G happened to touch at a single point, the construction would be completed immediately, and the point of tangency would itself be the point dividing the line into two equal parts.

The sixth of these will be Proposition I.11. It can be constructed easily by marking, on either side of the point, a segment equal to the given compass opening. Then, by means of the preceding proposition, bisect each of those segments. In this way, the two parts on either side of the point, taken together, will be equal to the given compass opening, and thus |p.29| by means of Proposition I.1, by constructing an equilateral triangle upon it according to the given compass opening, and drawing a line from the upper vertex to the given point, that line will be the perpendicular. The proof is perfectly clear from the first and the third of these propositions, since, according to the construction, two equilateral triangles are formed.

The seventh, eighth, and ninth of these propositions will be Propositions I.13, I.14, and I.15, which, because they are proved in the same way as by Theon, assume no propositions beyond those already proved, nor do they make use of the excluded postulate. It follows that they too have been proved according to our purpose.

Our tenth proposition will be: *Given two unequal lines issuing from the same point, to cut off from the greater a part equal to the lesser.* Let the two lines be AB and AC, of which AC is the greater.



To carry out the construction, by the fourth of these propositions I bisect the angle A by means of the indefinite line AE. Then, taking the point B as the centre, I draw a circle with the given compass opening, namely ED, and let it intersect the line AE at the point E. Next I draw the segment BE. Then, taking the point E as the centre, I draw, with the same compass opening, the circle GK, which intersects the line AC at the point F. Finally, I draw the segment EF, and I say that the construction has been completed; namely, that the line AB is equal to AF.

To prove this, I superimpose the triangle ABE upon the triangle AFE, the line AE remaining upon itself. Since the angle BAE is equal to the angle EAF, it is clear that AB will fall upon AF. If, therefore, the point B falls upon the point F, we have the desired result by the eighth common notion. But if the opponent should say that it reaches, not as far as F, but only as far as L, so that BE lies where EL is, I extend the line EL as far as G; and thus, since BE is already equal to EG, it would follow that EL is equal to EG—the part equal to the whole—which is impossible.

The same absurdity follows if one should maintain that B falls below F, as at H, so that EH comes to be EB. For then the part EK would be equal to the whole EH, which is impossible.

And thus |p.30| . . .